



Engineering the facilities advanced industry runs on.

EPCM, EPCC and Energy Facility Management for cleanroom-critical industries: semiconductor, data centre, EV battery, pharmaceutical and energy. Established 1994.



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Three parts. Every section is numbered, named and opens with a one-line summary.

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PART 01 OF 03

01

The Company

SECTIONS 01.1 TO 01.8

Established 1994 · 32 years in total facility solutions

450 people across 6 global offices

ISO 9001 · ISO 14001 · ISO 45001 certified



About IAQ

Who we are, what we build, and the models we deliver under.

Established 1994. A trailblazer in total facility solutions.

Engineering, procurement, construction and maintenance for the industries where contamination, uptime and safety decide the outcome, from semiconductor wafer fabs to gigafactories, data centres and pharmaceutical plants.

Vision

To be a regional facility solutions provider with engineering excellence that facilitates technological innovation and advancement in quality of life.

Mission

Providing innovative, sustainable facility and engineering solutions benefitting our clients and stakeholders, driven by our leadership, employees and partners globally.

Delivery Models

EPCM · EPCC · ENERGY FACILITY MANAGEMENT

Engineering and construction management, full EPCC from design through commissioning, and ongoing energy and facility management.

IAQ at a Glance

The company in six numbers, and the certifications that stand behind them.



VISUAL PLACEHOLDER

Corporate facility exterior, IAQ delivered project



32

YEARS EXPERIENCE



450

EMPLOYEES



200+

PROJECTS COMPLETED



1.5 mil m²

CLEANROOM BUILT-UP AREA



6

GLOBAL OFFICES



20 t / yr

CARBON FOOTPRINT REDUCED

CERTIFICATION & RECOGNITION

INTERTEK ISO 9001:2015

INTERTEK ISO 14001:2015

INTERTEK ISO 45001:2018

CIDB GRADE G7

PKK GRADE G7

HIGHWIRE SAFETY GOLD AWARD

MCIEA 2024 BUILDER OF THE YEAR

Milestones

Landmark projects from 1994 to 2026, grouped into five eras of growth. The full project list is in 02.11.



1994 – 2000

Foundation

1994
Established as a cleanroom specialist

2000
First decade of cleanroom delivery in Malaysia



2006 – 2009

Regional Expansion

2006
(EPCC) ST Microelectronics, Class 10K cleanroom

2008
(PCC) Western Digital PJ, Class 10 cleanroom

2009
(GC / D&B) MEMC Ipoh, Class 1 cleanroom



2013 – 2017

Landmark Plants

2013
(GC) KLCC DCC Plant, largest in Malaysia

2016
(EPCC) SilTerra FAB · (EPC) Infineon MKZ Class 1K

2017
(GC) Ain Medicare



2020 – 2022

Global Scale

2020
(GC) Vital Healthcare

2021
(GC / PCC) Robert Bosch · P Project

2022
(PCC) Microsoft Data Center · (EPCM / D&B) Soitec PR1A



2023 – 2026

Advanced Technology

2023
(GC / D&B) XFAB 40K Expansion

2025
(PCC) 160MW hyperscale data centre · (EPCC) ESCM Germany

2026
(D&B) Tata Dholera DF1 · (EPCM / HU) Micron MSH

Global Footprint

Where we operate, and the year we arrived in each market.



1994

Malaysia

Turnkey design, procurement, construction and commissioning for hi-tech clients. Selangor (HQ), Penang, Johor and Kuching.



2020

Sweden

Gigafactory construction for EV battery manufacture, part of Europe's clean and digital transition.



2023

Singapore

Design & build EPCM services to global semiconductor clients.



2025

Germany

EPCC services following semiconductor expansion in Europe.



2026

India · United States

D&B turnkey for India's first wafer fab, and US entry following localisation of advanced tech facilities.

ADDITIONAL OFFICES



Norway



China



France



Poland

Core Values

Six commitments that govern how every project is run. Not words, but working rules.



01

Safety First

Our services and works are carried out to the highest standard of safety and ethics.



02

Quality Consistency

We take pride in the quality of solutions delivered, to achieve maximum client satisfaction.



03

Honesty and Integrity

Our core code of conduct, fostering trust, accountability, professionalism and ethical practice.



04

Efficiency & Proficiency

Committed to maximising available resources to achieve the best result collectively.



05

Engineering Capabilities

Our way of working revolves around engineering principles, developing precise and sustainable solutions.



06

Pursuit of Excellence

Devoted to excellence in all our works, providing sustainable solutions to complex challenges.

Business Units

Three companies under one group, covering the full delivery chain.



VISUAL PLACEHOLDER

EPCC delivery on an active construction site



VISUAL PLACEHOLDER

Process critical utilities and tool installation inside a fab



VISUAL PLACEHOLDER

Energy plant and facility management operations

01

IAQ Solutions Sdn Bhd

ENGINEERING, PROCUREMENT, CONSTRUCTION & COMMISSIONING

EPCC from conception to operation, completion and maintenance. Every stage managed, from initial design through commissioning, so clients in hi-tech industries can bring their visions to life.

02

IAQ Utility Solutions Sdn Bhd

PROCESS CRITICAL UTILITIES & TOTAL TOOL INSTALL

EPCM partner for semiconductor manufacturing: process-critical utility infrastructure and total tool installation, bridging facility readiness and manufacturing start-up to accelerate fab ramp-up.

03

IAQ Energy Facility Management Sdn Bhd

ENERGY MANAGEMENT

Energy management solutions that optimise operations and reduce carbon footprint, keeping facilities running at the highest levels of efficiency and sustainability.

Scope of Services

Twelve packages, self-performed and integrated under one contract.



01

Cleanroom System

ISO 3 to ISO 8 · Class 1 to Class 100K



02

Air Conditioning System

ACMV, make-up air, recirculation



03

Process Utilities

CDA, PCW, PV



04

Process Exhaust System

Scrubbed, heat and general exhaust



05

Chemical & Gas Delivery

Specialty and bulk gases, chemical delivery



06

Ultra Pure Water

UPW generation and distribution



07

Waste Water Treatment

Industrial effluent and recovery



08

Fire Protection System

Detection, suppression, compliance



09

HT & LV Electrical System

Up to 33kV substations



10

Civil, Structural & Architectural

Greenfield and brownfield CSA



11

Facility Monitoring & Control

FMCS, BMS, PA



12

Tools Hookup

Progressive tool install and hookup

Industry Focus

Seven sectors, one discipline: contamination control under code and class.



01

Semiconductor

Wafer fab, advanced packaging, test & assembly



02

Data Centre

Hyperscale cooling, security, uptime



03

EV Battery

Dry rooms, moisture control, explosion proofing



04

District Cooling & Heating

Urban energy plant and distribution



05

Photovoltaics

Cell and module lines, toxic material handling



06

Pharmaceuticals & Hospitals

GMP grades, regulated environments



07

Food & Beverage

Hygienic design, quality and safety standards



+

Your facility next

Total facility solutions, end to end

PART 02 OF 03

02

Project References

SECTIONS 02.1 TO 02.11

200+ projects completed across three regions

Over 1.5 million m² of cleanroom built

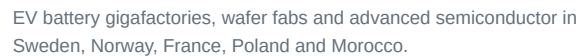
ISO 3 to ISO 8 · Class 1 to Class 100K



How the reference book breaks down by region, and which sections cover what.



PROJECTS COMPLETED



CLEANROOM BUILT



GC · EPCC · EPCM · PCC · Tool Hookup

Semiconductor · Malaysia

Wafer fab, advanced packaging, test and assembly. Further Malaysian projects are listed in 02.11.



KULIM, KEDAH · GREENFIELD WAFER FAB

Infineon Kulim Wafer Fab 3

CLEANROOM	ISO 4 · 5 · 6 · 7 (CLASS 10 · 100 · 1K · 10K)
SCOPE	PCC for cleanroom and mechanical works
DESCRIPTION	General contractor for WP06, KLM3 Expansion, 35,000 m ² cleanroom area



ADVANCED PACKAGING · 75,000 M²

P Project

CLEANROOM	ISO 4 TO 7 (CLASS 10 TO 10K)
SCOPE	Design & build GC: CSA, MEP, process utilities, chemical & gases system, waste treatment
DESCRIPTION	Progressive-built EPCM contract model for the client's advanced packaging facility

Semiconductor · Singapore

Design & build EPCM in Singapore since 2023.



PHOTOMASK FACILITY

Teksend Photomask

CLEANROOM	CLASS 1 TO CLASS 10K
SCOPE	EPCC for cleanroom, ACMV, electrical, process utility, BMS, PA, WWT, UPW, tool hookup
DESCRIPTION	Addition and alteration to an existing two-storey building with cleanroom facilities and ancillary office



PROGRESSIVE TOOL HOOKUP

Micron MSH Tool HU

SCOPE	Progressive tool hook up
DESCRIPTION	2,000 progressive tool hookup services for MSH

Data Centre

Hyperscale and colocation, 9.6 MW to 160 MW.

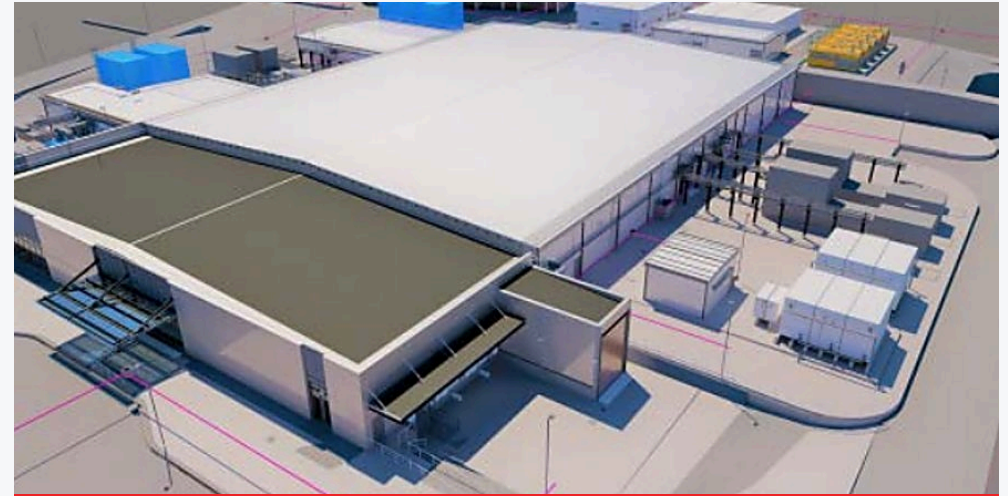


CLIENT CONFIDENTIAL · 160 MW

Hyperscale Data Centre

SCOPE Construction, completion, testing & commissioning for mechanical packages 1 & 2

DESCRIPTION Office, data halls and mechanical utility building



KUL03 PHASE 1 · 9.6 MW

Microsoft DTC-KUL 03

SCOPE PCC for CHW and CW piping works

DESCRIPTION 9.6 MW data centre

EV Battery · Europe

Dry rooms for the gigafactories: moisture control and contamination control at scale.



SKELLEFTEÅ, SWEDEN · 62,000 M²

Northvolt AB Gigafactory

ENVIRONMENT	DRY ROOM
SCOPE	Cleanroom and dry room architecture works
DESCRIPTION	Europe gigafactory for lithium-ion battery manufacture



NORWAY · BATTERY COAST · 8,600 M²

Morrow Batteries

ENVIRONMENT	DRY ROOM
SCOPE	Cleanroom and dry room architectural system, ACMV
DESCRIPTION	Norway's Battery Coast project

Semiconductor · Europe & Morocco

Following the fabs into Europe: advanced-node cleanrooms and progressive hookup.



CLIENT CONFIDENTIAL · WAFER FAB

28/22nm CMOS · 16/12nm FinFET Plant

CLEANROOM CLASS 1 TO CLASS 10K

SCOPE Cleanroom & mechanical general contractor

DESCRIPTION Design, supply, installation & commissioning of cleanroom facilities



CASABLANCA, MOROCCO · 60,000 M²

STMicroelectronics

CLEANROOM ISO 5 (CLASS 100)

SCOPE Cleanroom, mechanical, electrical, process utilities and hookup works

Semiconductor & Display · China

EPCC across semiconductor, display and precision manufacturing.



HANGZHOU · 23,400 M²

Ferrotec Semiconductor

CLEANROOM ISO 4 · 5 (CLASS 10 & 100)

SCOPE EPCC for cleanroom, mechanical, electrical and plumbing works and hookup



WUXI · 8,000 M²

Infineon Technologies

CLEANROOM ISO 5 (CLASS 100)

SCOPE EPCC for cleanroom, mechanical, electrical and plumbing works and hookup

Photovoltaics

Solar cell and module manufacturing, including toxic material and waste handling.



KULIM, MALAYSIA

First Solar Malaysia

SCOPE PCC for mechanical & electrical, KMW building; CSA and M&E for forming gas plant; M&E for SCO1



CHINA · 57,000 M²

Chint Solar Technology

CLEANROOM ISO 8 (CLASS 100K)

SCOPE Cleanroom system, ACMV, process utilities and tools hookup

Pharmaceutical & Medical

GMP grades and regulated environments, from parenteral facilities to medical device plants.



GREENFIELD · MEDICAL DEVICE

Insulet

CLEANROOM ISO 8 (CLASS 100K)

SCOPE Design and build mechanical, cleanroom & electrical package



PUCHONG, SELANGOR

Pharmaniaga

CLEANROOM ISO 5 · 7 · 8 (CLASS 100 · 10K · 100K)

SCOPE Cleanroom system and ACMV for a small volume parenteral facility

District Cooling & Energy

Plant, chillers and co-generation for urban and industrial energy.



GAS DISTRICT COOLING (M) SDN BHD

KLCC District Cooling Plant

DESCRIPTION LARGEST DISTRICT COOLING CENTRE IN MALAYSIA

SCOPE ACMV, electrical & instrumentation, fire protection, process utility, plumbing & sanitary, water treatment, CSA



PICC PLANT, PUTRAJAYA

GDC Putrajaya

SCOPE EPCC of plant electrification, chiller replacement and associated works at Gas District Cooling Putrajaya

Reference Index

A selection from 200+ completed projects. The full reference list is available on request.

MALAYSIA · SEMICONDUCTOR

XFAB Kuching 40K Expansion, 40,000 m²

SOITEC PR1A Expansion, 45,000 m²

Robert Bosch Testing Manufacturing Plant

Wafer Fab Facility & Expansion, FAB1E A/B

Texas Instruments, Melaka, 32,000 m²

Infineon Technologies, Melaka, Block 8

MEMC Ipoh, Class 1 cleanroom

STMicroelectronics, Johor

Western Digital (M), Petaling Jaya

WD Media (M), Penang, MA 28 & MA 29

Flextronics Shah Alam

MALAYSIA · OTHER SECTORS & SINGAPORE

AUO Sunpower, Melaka, Fab 3A

Vital Healthcare, medical device plant

Ain Medicare, pharmaceutical building

M Project, food flavouring plant, 40,000 m²

P Project, MEP works, mega lab retrofit

Rapid Petronas Pengerang Co-gen

Pagoh Education Hub DCS

Rapid MCD chemical plant

Matrix (M) Sdn Bhd

T Hasegawa, spray dryer & microwave tunnel

Elegant Aura (M) Sdn Bhd

Caterpillar Asia, 7T office renovation

Micron F10A & F10NX, Singapore

National University of Singapore, tissue culture lab

EUROPE & MOROCCO

ACC Phase 1, France, 35,000 m²

Envision, France

Soitec Semiconductor, Paris, 11,000 m²

Xfab Semiconductor, Paris

Nemotek, Rabat, Morocco

Kimoto, Poland

Sumika, Poland

CHINA

SMIC, Ningbo, 22,000 m²

Maxtor Technology, Suzhou, 95,000 m²

All-Cent RF Technology, Wuxi, 25,600 m²

Kunshan Visionox Display, 27,000 m²

GS Magicdrive Inc., 68,000 m²

Shantou Goworld Display

M-Flex (Suzhou), 25,000 m²

MMI Industries (Wuxi)

Huawei Technologies Co., Ltd

Shanghai Roche Pharmaceuticals

Suzhou Ascentage Pharma, 25,800 m²

Jiangsu GenScript Biotech

Mindray Group Co., Ltd

Unilever · GE · Dow · Rohm and Haas R&D centres

PART 03 OF 03

03

Safety, Quality & ESG

SECTIONS 03.1 TO 03.4

- Highwire Gold safety award
- ISO 45001:2018 occupational health & safety
- 20 tons per annum carbon footprint reduction



VISUAL PLACEHOLDER

Site safety in practice: crew in full PPE, toolbox briefing or EHS walkdown on an IAQ site

ESG Commitments

Three pillars, applied to operations and to every project we take on.



E

Environmental Responsibility

Minimising environmental footprint through sustainable design and construction. Energy-efficient solutions and eco-friendly technologies, prioritised at design stage.



S

Social Accountability

The well-being and safety of employees and communities first. A diverse, inclusive work environment and ethical labour practices.



G

Corporate Governance

Transparency, accountability and ethical decision-making, with the highest standards of compliance, building trust with clients, partners and stakeholders.

Safety & Recognition

How our standards are independently verified and recognised.

Judged on the things that actually matter.

MCIEA award recipients are evaluated across company performance, project management, technical expertise, innovation, quality, safety and sustainability. The same criteria our clients audit us against.

Builder of the Year 2024

Malaysian Construction Industry Excellence Awards (MCIEA), awarded by CIDB.

Highwire Gold

Gold-rated contractor safety performance.

ISO 45001:2018

Occupational health and safety management, certified by Intertek.



Closing

Your Total Facility Solutions Provider.

We infuse safety, quality, efficiency and excellence into every project.

From engineering design and project management to construction and maintenance, across semiconductor manufacturing, clean energy and every environment in between. IAQ is your trusted partner for total facility solutions.

Contacts

For inquiry and business opportunity.



YOUR TOTAL FACILITY SOLUTIONS PROVIDER

Engineering, procurement, construction and energy management for the industries where contamination, uptime and safety decide the outcome. Established 1994.

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